

DS	Session	Linked List	Question Information	Level 1	Challenge 34
Question description			Question No:4		
Problem Description:					
For Engineering Second -year students, the instructor is organising a surprise test.					
He's given you a new node is added before the given node of the given Linked List.					
For example if the given Linked List is 5->10->15->20->25 and					
we add an item 30 before node 15, then the Linked List becomes 5->10->30->15->20->25.					
Since a Linked List is typically represented by the head of it, we have to traverse the list till node and then insert the node.					
Constraints:					
1 < arr <100					
Input Format					
First line contains the number of datas- N.					
Second line contains N integers(the given linked list).					
Third line contains the node P before which the node to be inserted.					
Fourth line contain the node X to be inserted.					
Output Format					
case 1 (node P found in Linked list) :					

Display the final Linked List.

case 2(node P not found in Linked list) :

Print Node not found!

Display the Linked list.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
115 210 116 110
116
17
```

EXPECTED OUTPUT

```
Linked List:->115->210->17->116->110
```

Test Case 2

INPUT (STDIN)

```
4
15 20 16 10
25
17
```

EXPECTED OUTPUT

```
Node not found!
Linked List:->15->20->16->10
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
node *next;
```

Test Case 3

KEYWORD

```
*start
```

Test Case 4

KEYWORD

```
p2->next=p1;
```

Test Case 5

KEYWORD

```
void display()
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct node
5  {
6      int data;
7      struct node *next;
8  };
9
10 struct node *head = NULL;
11
12 void insertEnd(int data)
13 {
14     struct node *newNode;
15     struct node *temp;
16
17     newNode = (struct node *)malloc(sizeof(struct node));
18
19     newNode->data = data;
20     newNode->next = NULL;
21
22     if (head == NULL)
23     {
24         head = newNode;
25         return;
26     }
27
28     temp = head;
```

```
29
30     while (temp->next != NULL)
31     {
32         temp = temp->next;
33     }
34
35     temp->next = newNode;
36 }
37
38 int insertBefore(int p, int x)
39 {
40     struct node *newNode;
41     struct node *temp;
42
43     /* Insert before the first node */
44     if (head != NULL && head->data == p)
45     {
46         newNode = (struct node *)malloc(sizeof(struct node));
47
48         newNode->data = x;
49         newNode->next = head;
50         head = newNode;
51
52         return 1;
53     }
```

```
55     temp = head;
56
57     while (temp != NULL && temp->next != NULL)
58     {
59         if (temp->next->data == p)
60         {
61             newNode = (struct node *)malloc(sizeof(struct node));
62
63             newNode->data = x;
64             newNode->next = temp->next;
65             temp->next = newNode;
66
67             return 1;
68         }
69
70         temp = temp->next;
71     }
72
73     return 0;
74 }
75
76 void display()
77 {
78     struct node *temp = head;
```

```
80     printf("Linked List:");
81
82     while (temp != NULL)
83     {
84         printf("->%d", temp->data);
85         temp = temp->next;
86     }
87
88     printf("\n");
89 }
90
91 int main()
92 {
93     int n;
94     int i;
95     int value;
96     int p;
97     int x;
98     int found;
99
100     scanf("%d", &n);
101
102     for (i = 0; i < n; i++)
103     {
104         scanf("%d", &value);
105         insertEnd(value);
106     }
```

```

106     }
107
108     scanf("%d", &p);
109     scanf("%d", &x);
110
111     found = insertBefore(p, x);
112
113     if (found == 0)
114     {
115         printf("Node not found!\n");
116     }
117
118     display();
119
120     return 0;
121 }

```

Output:

Test Case-1:

```

4
115 210 116 110
116
17
Linked List:->115->210->17->116->110
PS C:\Users\kesha\OneDrive\Documents\3sem\C>

```

Test Case-2:

```

4
15 20 16 10
25
17
Node not found!
Linked List:->15->20->16->10
PS C:\Users\kesha\OneDrive\Documents\3sem\C>

```

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